Math 150A Section 8 and 16
Fall 2025
Final Exam
May 13, 2025
Time Limit: 1 hour and 50 Minutes

Name (Print): ______________________________

Student ID: ______________________________

This exam contains 9 pages (including this cover page) and 8 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.

You may *not* use your books or notes on this exam. However, you may use a single, handwritten, one-sided notesheet and a *basic* calculator.

You are required to show your work on each problem on this exam, unless otherwise stated. The following rules apply:

- **Organize your work**, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit**. A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit. This especially applies to limit calculations.
- If you need more space, use the back of the pages; clearly indicate when you have done this.
- **Box Your Answer** where appropriate, in order to clearly indicate what you consider the answer to the question to be.
- Unless otherwise stated, all angles are in **radians**

Do not write in the table to the right.

Problem	Points	Score
1	20	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
8	10	
Total:	90	

1. (20 points) **TRUE or FALSE!** If the statement is true, write TRUE (all caps, all spelled out). Otherwise write FALSE (all caps, all spelled out).

   (a) the function $f(x) = e^x$ has a critical point

   (b) x^2 has an inflection point at $x = 0$

   (c) if $f(x)$ is continuous at $x = 2$, then it is differentiable at $x = 2$

   (d) if $f'(1) = 2$ and $g'(1) = 3$, then the derivative of $f(g(x))$ at $x = 1$ is 6

   (e) $\tan(\tan^{-1}(10)) = 10$

   (f) $\int_0^1 2f(x)dx = 2\int_0^1 f(x)dx$

   (g) $\frac{d}{dx}e^{2x} = 2e^{2x}$

   (h) if $f(x)$ is differentiable and $f'(x) = 0$, then $f(x)$ is a constant

   (i) the function $e^x + x$ has a root in the interval $[0, 2]$

   (j) a function can only have one vertical asymptote

2. (10 points) Calculate the following. Carefully show you work.

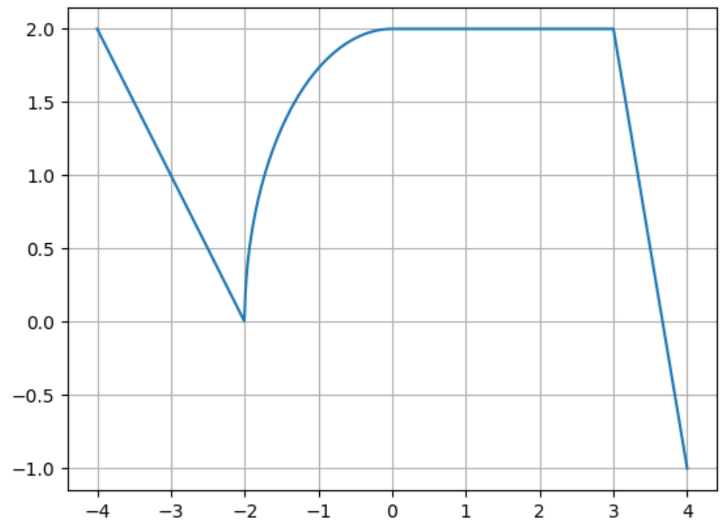
   (a) $\lim_{x\to 2} \frac{x^2-5x+6}{x^2-4}$

   (b) $\lim_{x\to\infty} \sqrt{x^2+1} - \sqrt{x^2+2x}$

   (c) $\lim_{x\to 0} \frac{e^{2x}-1}{3x}$

   (d) $\lim_{x\to\infty}(1+5x)^{1/x}$

3. (10 points) The following is a graph of the **derivative** $f'(x)$ of a function $f(x)$.



(a) List all intervals where $f(x)$ is concave down

(b) List all critical points of $f(x)$ and classify them as local max, local min or neither

(c) What is $\lim_{x\to -3} \frac{f'(x)}{x+3}$?

(d) If $f(-4) = 1$, what is the value of $f(3)$? Assume $f(x) = \sqrt{4 - x^2}$ for $-2 \le x \le 0$

4. (10 points)

   1. Find all critical points of the function $f(x) = \frac{e^x - 1}{e^{2x} + 1}$

   2. If $f(2) = 0.75$ and $f'(2) = -0.25$, use the linearization of $f(x)$ at $x = 2$ to estimate $f(1.9)$.

5. (10 points) Find the derivative of each of the following functions.

(a) $x \tan^{-1}(x^2)$

(b) $\frac{\cos(x)}{x+2^x}$

(c) x^x

6. (10 points) Consider a rectangular box with a square base, an open top, and a volume of 216 cubic inches. What dimensions minimize the surface area of the box? What is the minimum surface area?

7. (10 points) Find the area of the region bounded by $y = -x^2 + 10$ and $y = (x-2)^2$.

8. (10 points) Compute each of the following integrals. Show all work. Lack of work results in no credit!!

   (a) $\int_1^8 \frac{3+t}{t} dt$

   (b) $\int 2xe^{x^2} dx$

   (c) $\int \frac{e^s}{1+e^{2s}} ds$

   (d) $\int_0^3 (x^3 + 3^x) dx$